

Sovereign by 2031

Indigenous Cloud • Indigenous Miniature Satellites

A five-year blueprint to own the compute, the network and the sky above our platform — built only for Build Objects's one lakh customers, never for sale.

One platform. Three products. Zero rented rails.

WHAT WE ARE BUILDING



Build Objects Studio

AI architect — sanction-grade designs, 3D + 4K renders



Build Objects Store

15,000+ SKUs — cement to chandeliers, BOQ to cart



Build Objects Construct

Contractors on escrow, milestone-tracked sites

One BIM spine — the design file drives every rupee spent after it.

WHY OWN THE INFRASTRUCTURE



Cost control

GPU rendering is our biggest recurring bill



Sovereignty

Designs, payments, homes — data stays in India, with us



Uptime

10,000 concurrent designers; sites that can't go dark

Vertical integration is the moat competitors can't rent.

The thesis: *whoever owns the design owns the transaction — whoever owns the rails owns the margin, the latency and the risk.*

SERVE OURSELVES ONLY — NOTHING IS FOR RESALE

The load we must never drop

1,00,000

customers on platform

10,000

house designs / year

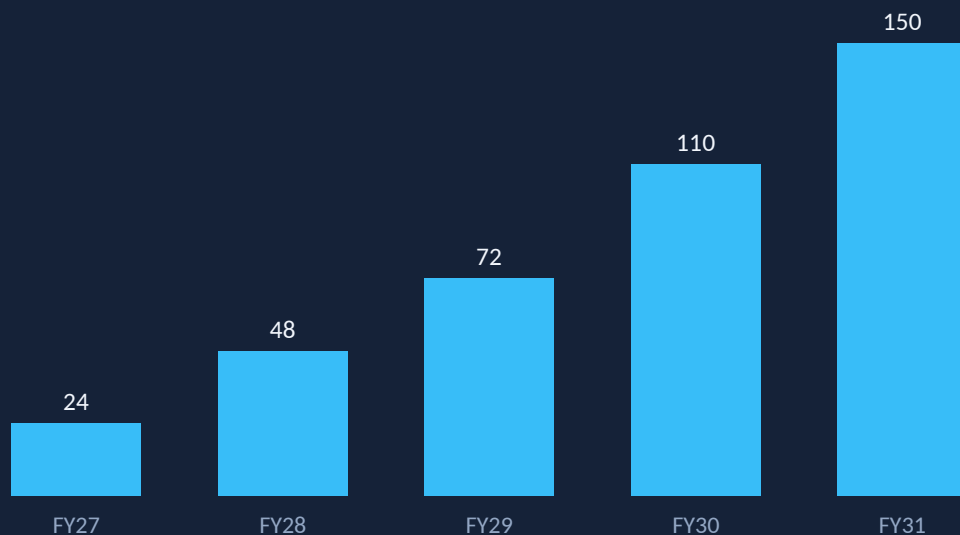
10,000

concurrent users at peak

99.95%

uptime target

RENDER GPU FLEET (SUSTAINED CARDS)



WHAT 10,000 CONCURRENT MEANS



10-20k requests/sec at the edge

≥ 90% absorbed by CDN cache



~1-2k requests/sec at origin

30-40 stateless app pods, autoscaled



2,000+ renders / day

queue-scheduled; spikes = longer ETA, never a crash



150 TB → 1.5 PB objects

renders + 4K walkthroughs; DB stays < 2 TB

Own the stack, layer by layer



L3 — THE SKY

12 miniature satellites + leased LEO broadband



L2 — THE CORE

Own 1 MW data center + DR + edge cache



L1 — THE CLOUD

Build Objects private cloud — K8s, GPU farm, object store

2026 → 2031

Rented cloud

→ Private cloud

Public internet

→ Private SD-WAN

Others' satellites

→ Our constellation

Vendor SOC

→ Our 24x7 SOC

One control plane, one security operations center, one bill — ours.

The Build Objects private cloud



Users → CDN edge

90%+ cache hit — renders, catalogs, galleries



Kubernetes app plane

Next.js · NestJS · design engines · edit-agent



Data plane

PostgreSQL + PostGIS · Redis · Meilisearch



GPU render farm

L40S-class path tracing · priority queues



Object store + archive

S3-compatible · lifecycle tiering · CDN-fronted



Two Indian regions

Hyderabad primary + DR region · RPO 15 min, RTO 4 hr



Open-source core

K8s, Postgres, MinIO/Ceph — zero lock-in, export path always live



Keys stay home

Encryption keys in our HSMs — never a foreign cloud's custody



Burst valve

Public-cloud GPUs only for spikes — never for the baseline

DESIGN LAW — cloud-neutral: any workload can leave any vendor in 90 days

Rent, partner or own the servers?

5-YEAR CUMULATIVE CASH OUT (₹ CRORE, EST.)



Own the GPUs — always

Payback < 2 years above 60% utilisation



Rent the building — for now

Colo until FY29; our IT racks inside it



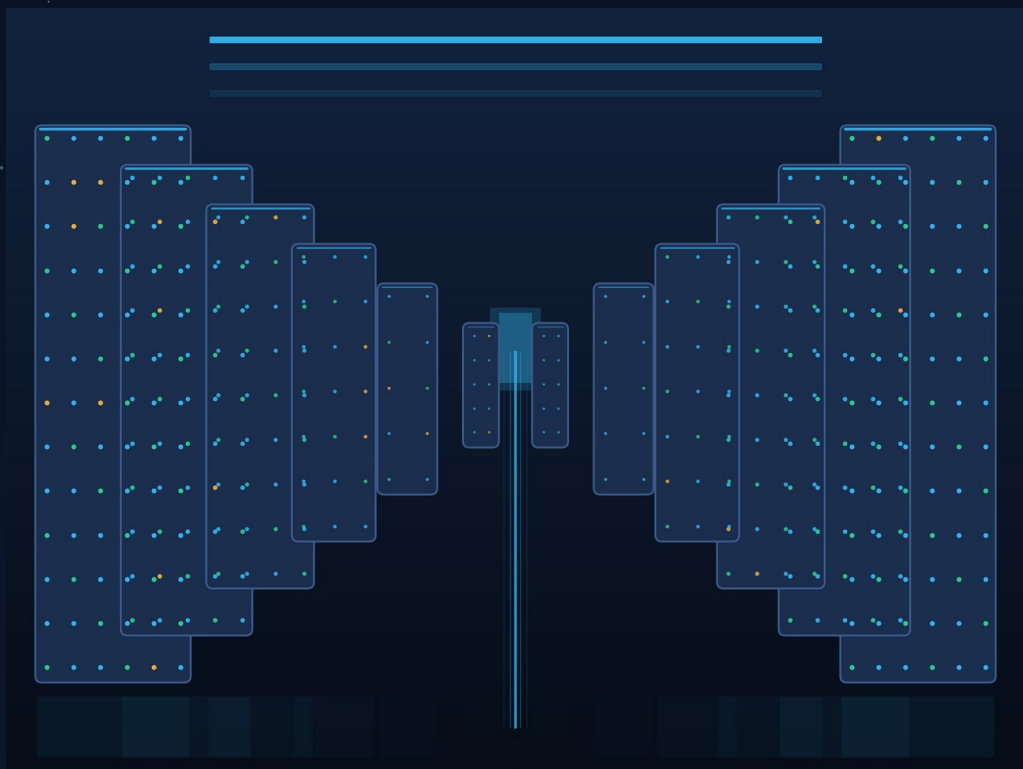
Burst to cloud — forever

Festival spikes rent GPUs by the hour

+₹49 Cr 5-yr premium of full ownership —
bought back after FY31 at ~₹7 Cr/yr savings

VERDICT — hybrid: own compute, lease shells, burst up

Land: our 1 MW sovereign core



HYDERABAD

DUAL GRID

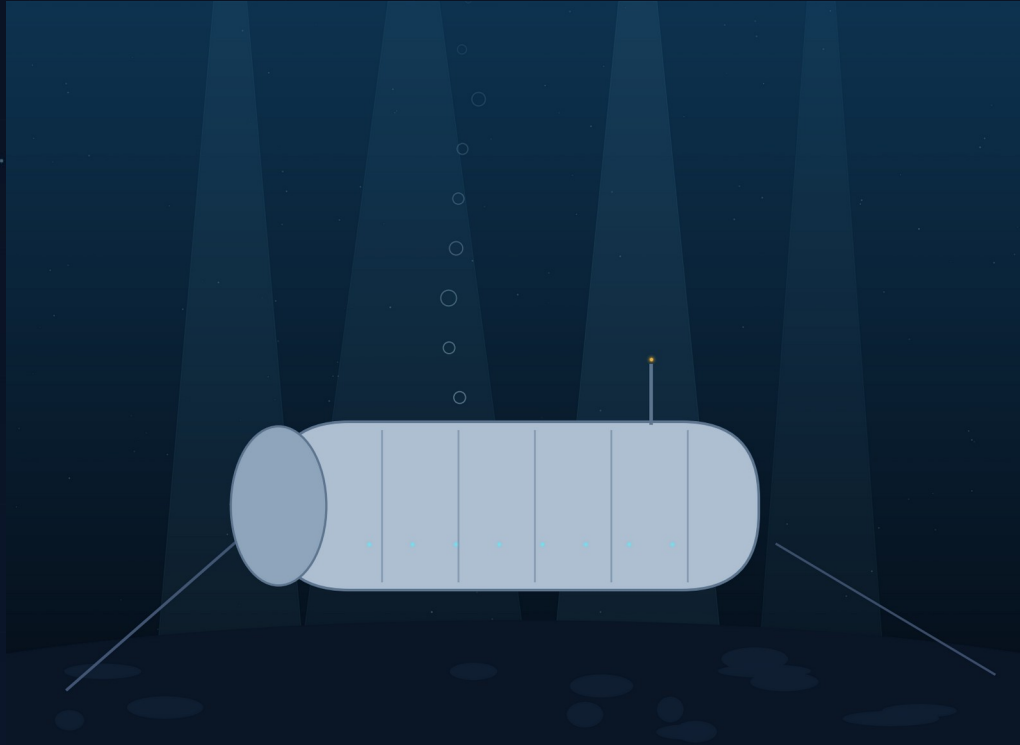
DUAL FIBRE

WHAT IT COSTS (EST.)	
Build — shell, power, cooling	₹50-70 Cr / MW
IT fit-out — servers, GPUs, network	₹40-55 Cr
Run — power, people, spares	₹8-12 Cr / yr
Timeline — design to live	18-24 months

Tier III	N+1	PUE ≤ 1.4	Liquid-ready
concurrently maintainable	power & cooling	with warm-water option	for next-gen GPUs

DECISION — site selection FY28 · ground-break FY29 · live FY30

Sea: real technology, not yet our build



PUE ~1.1 — SEA-COOLED

ZERO LAND, ZERO WATER



Proven reliable

Microsoft's subsea pods: 1/8th the failure rate of land — project still shelved in 2024



Now commercial — in China

First commercial subsea DC live off Shanghai, Oct 2025 — offshore-wind powered, ~90% cooling energy cut



India has no rulebook

No CRZ / marine framework for subsea compute; permits, cables, insurance all unmapped

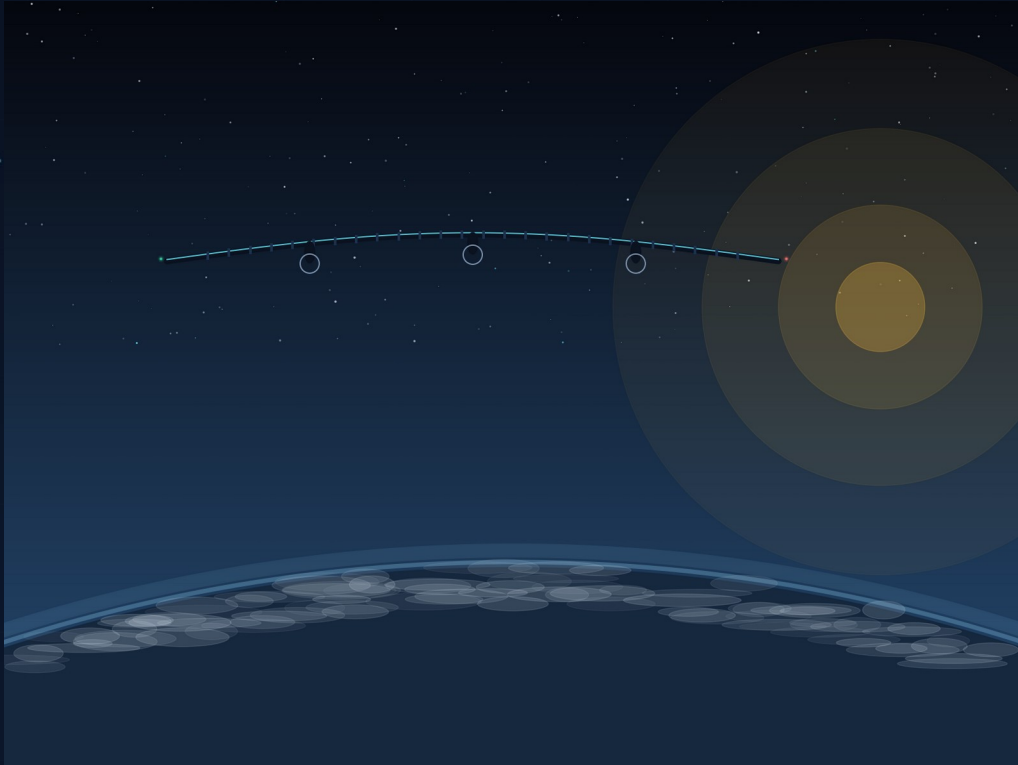


Our economics

Prototype ≈ ₹80–150 Cr — beats land only at coastal scale, after the rules exist

VERDICT — WATCH & PARTNER: joint feasibility study FY30, zero capex before rules

Sky: a relay layer, not a data center



20 KM · SOLAR HAPS

WEEKS ALOFT, KW POWER



The record is real

67-day continuous stratospheric flight (Zephyr, Apr 2025); first services target 2026



India is flying too

NAL's solar HAP prototype in test campaigns; HAL developing CATS warfighter ecosystem



Physics says no servers

A HAPS lifts kilowatts, not megawatts — compute payloads don't close; it's a flying cell tower



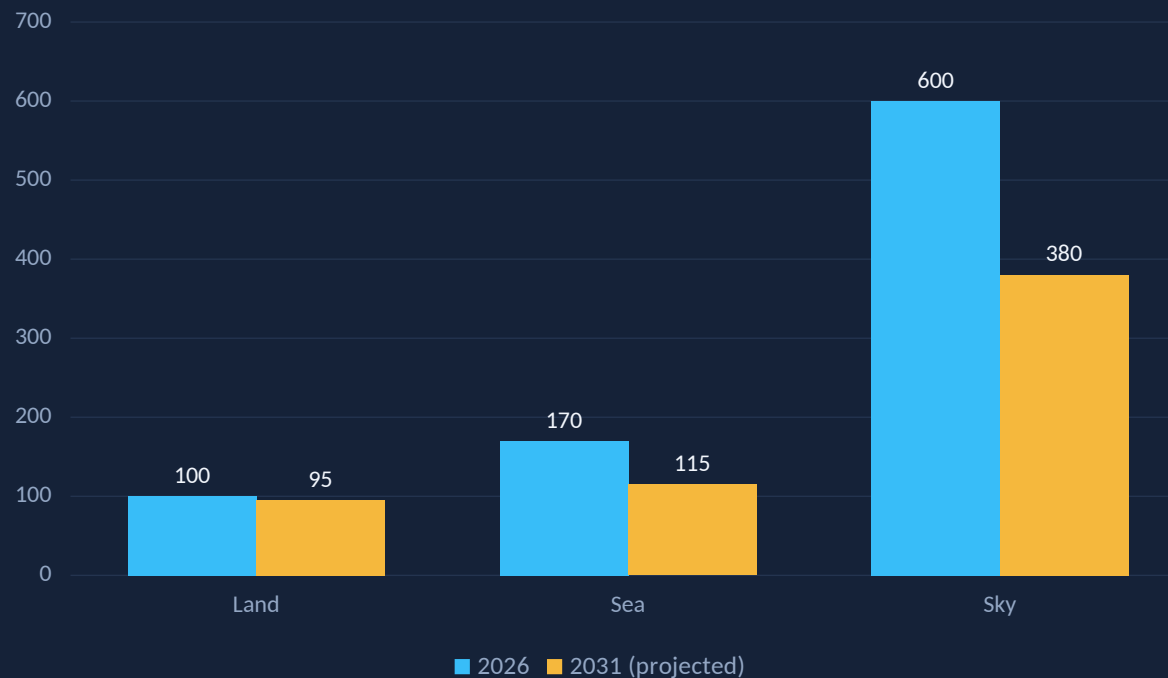
Our use case

Disaster-relay + temporary site backhaul over remote projects — rented by the week

VERDICT — NO COMPUTE ALOFT: rent HAPS relay trials FY30+, own nothing that flies in air

Land wins this decade — and we'll watch the water

COST INDEX PER MW DEPLOYED (LAND 2026 = 100, EST.)



Maturity

Land TRL-9 · Sea first-commercial · Sky pre-commercial

Cooling & PUE

Sea best (~1.1) · Land 1.3–1.5 · Sky n/a

Rules in India

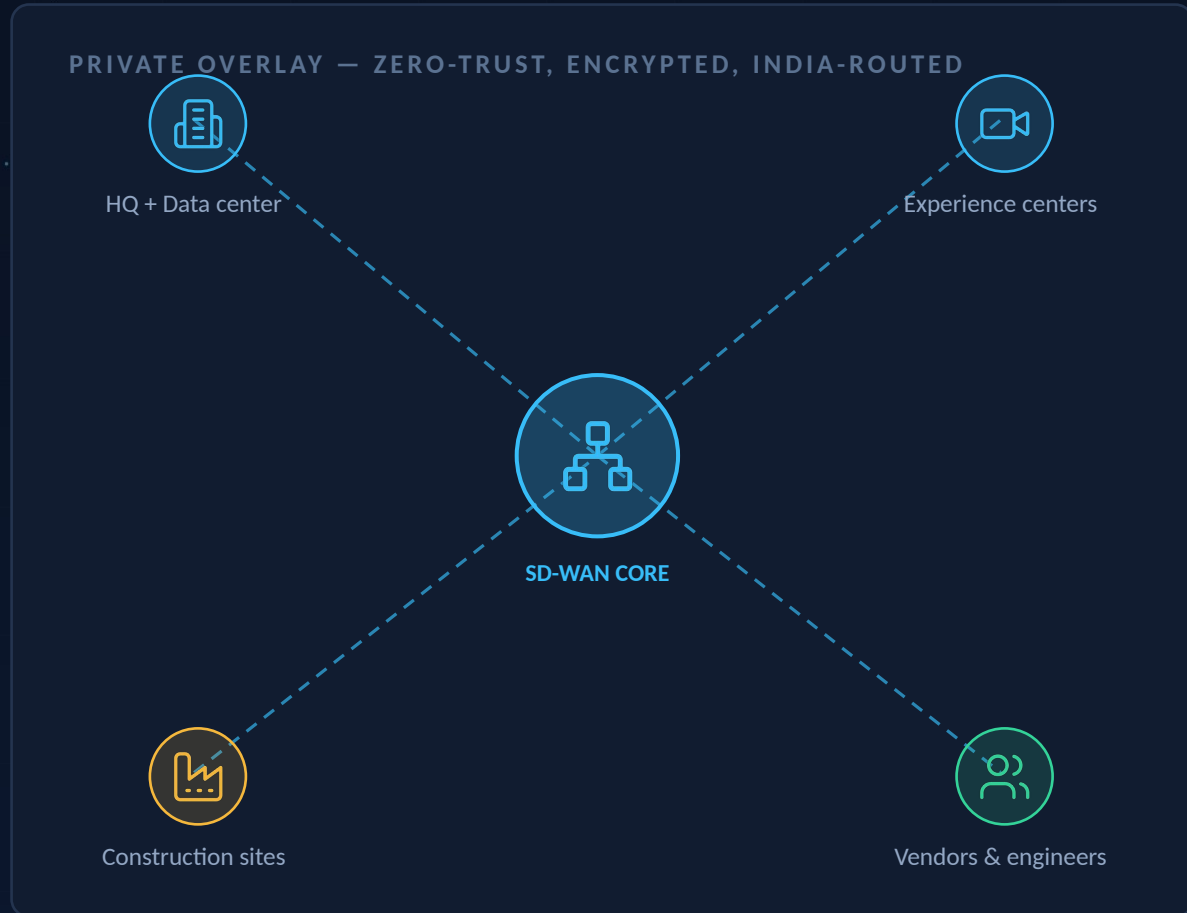
Land clear · Sea absent · Sky partial (DGCA)

Fix & maintain

Land hours · Sea weather-window · Sky landing cycles

BUILD on land · STUDY the sea · RENT the sky

Every site online, three ways down



Path 1 — Fibre

Primary at every fixed site; dual entry at DC



Path 2 — 4G/5G

Instant-on for new construction sites



Path 3 — Satellite

Leased LEO + our constellation; no-signal villages covered



One overlay

SD-WAN + zero-trust; site joins in a day, fails over in seconds

TARGET — 200+ sites by FY31 • 99.9% each • zero dark days

Buy the sky first — then build our corner of it

GEO

36,000 km • ~600 ms

Broadcast, rural VSAT

MEO

8,000 km • ~150 ms

Trunk backhaul

LEO



550 km • 25–50 ms

Interactive apps — our fit

WHO CAN SELL US CAPACITY IN INDIA (STATUS AUG 2026)

Starlink

GMPCS + IN-SPACE cleared; ~₹3,000/mo expected, terminal ₹25–35k

Jio-SES

GEO+MEO enterprise; Indian gateways live

Eutelsat OneWeb

LEO enterprise via Bharti; India gateways

Captive VSAT CUG

Our own licensed hub — ₹10k/terminal/yr fee

STRATEGY — lease LEO broadband for sites & failover • reserve our own build for the one job leasing can't do: captive telemetry

What miniature satellites can honestly do

CAPABILITY — TRUTH TABLE



NB-IoT telemetry

Site sensors, trackers, meters — few bytes, everywhere



Messaging & alerts

Milestone pings, SOS, fleet — store-and-forward



Scheduled data drops

Site photos, drone survey packets on each pass



Thin broadband

Mbps-class only with large sats + big gateways



Full broadband

Needs 100s of satellites & ₹8,000+ Cr — Starlink's game, not ours

THE CLASSES (BUILD COST / SAT, EST.)

1U–3U CubeSat

1–5 kg · student & demo missions

₹1–4 Cr

6U–12U nanosat

8–25 kg · our class — NB-IoT payloads

₹6–12 Cr

50–100 kg microsat

50–100 kg · thin comms, EO

₹15–35 Cr

150–300 kg smallsat

150+ kg · broadband-lite constellations

₹40–90 Cr

Bus platforms now indigenous: Dhruva · Astrome · Azista (IN-SPACe programme, 2026)

The Build Objects constellation — 12 birds by FY31



S1 · FY28-29 — Demonstrator

2× 6U NB-IoT sats · SSLV / rideshare · ground via GSaaS

S2 · FY30 — First ring

6 sats · own S-band gateway, Hyderabad · live site telemetry

S3 · FY31 — Full ring

12 sats · ~90-min revisit India · 2 gateways + mission NOC

550 KM SSO ORBIT

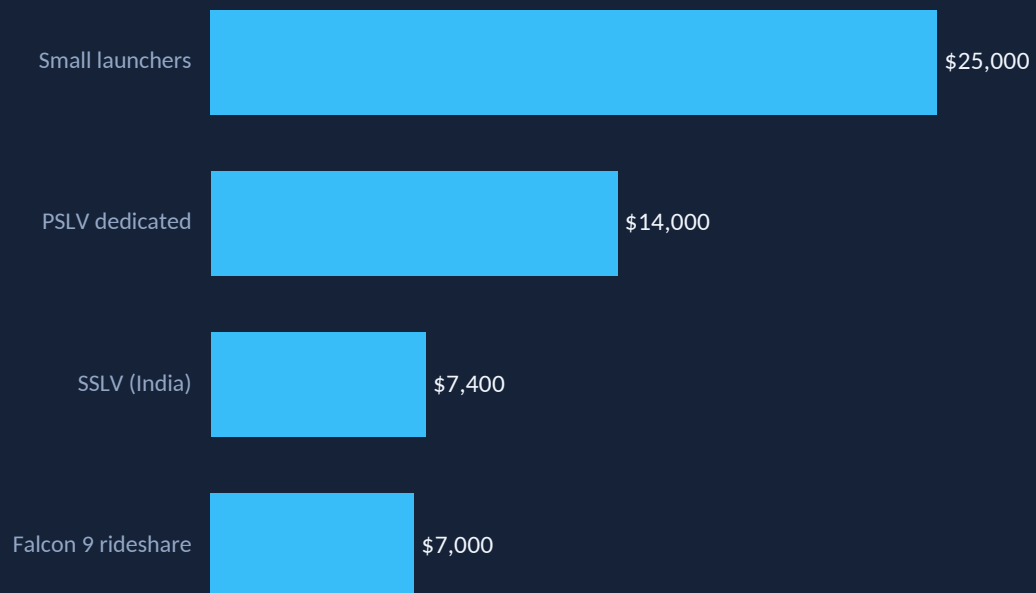
S-BAND NB-IoT (NTN)

STORE-AND-FORWARD

Built with, not from scratch: indigenous buses (Dhruva / Astrome / Azista), our payload spec, ISRO-ecosystem launch, Indian ground partners.

What the sky costs

LAUNCH PRICE PER KG TO ORBIT (USD, 2026)



Demonstrator (2 sats, FY28-29)

₹25-35 Cr

12-sat constellation, all-in

₹180-260 Cr

— build ₹96-144 · launch ₹20-35 · ground ₹25-35 · 3-yr ops ₹40-55

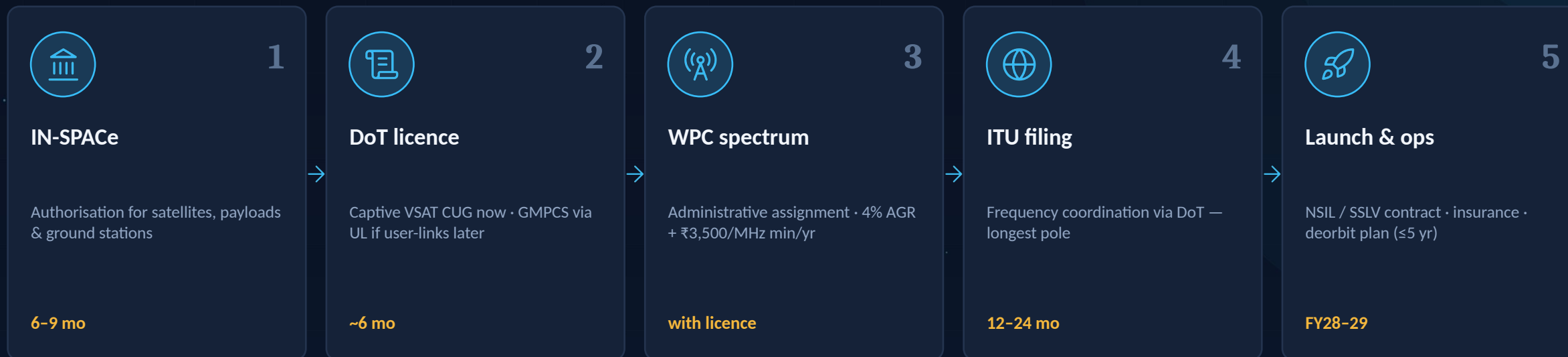
Leased-only alternative (per yr)

₹3-4 Cr

The honest math: owning never beats leasing on ₹/bit. We buy certainty: a layer nobody can price-hike, geo-block or switch off.

GATE IT — constellation go/no-go sits at FY29, after the demo flies

The regulatory runway — start filing now



WHAT WE ACTUALLY BUY

- ✓ S-band NB-IoT payload spectrum (NTN)
- ✓ Ku/Ka gateway feeder links
- ✓ S/UHF TT&C command links
- ✓ WPC ETA for every terminal type

FEES ARE SMALL, TIME IS NOT — file IN-SPACe + ITU paperwork in FY27, fly in FY29

Six rules we'll push to change



Captive-constellation licence

A simple own-use class for private constellations — like captive VSAT, not full GMPCS



120-day single window

IN-SPACe + DoT + WPC as one time-bound approval, not three queues



10-year spectrum terms

5-year assignments can't finance 7-year satellite capex



Guaranteed SSLV cadence

Published rideshare calendar so Indian constellations plan launches like freight



Marine DC framework

CRZ / offshore rules for subsea & floating data centers — today there are none



PLI for space + DC gear

Extend production incentives to satellite buses, gateways, HSMs, cooling

Route: IN-SPACe consultations · TRAI open houses · industry bodies (ISpA) — as a builder with skin in the game, not a lobbyist.

Sovereign means secure — or it means nothing



Data residency

Every byte — designs, payments, homes — stored and processed in India



Key custody

Encryption keys in our HSMs; no foreign cloud ever holds them



Zero-trust

MFA everywhere, least privilege, segmented networks, signed firmware



24x7 SOC + NOC

One pane for cloud, DC, network and satellites; forensic-ready logging



Unkillable backups

Immutable + offline copies; ransomware drills twice a year



Compliance built-in

DPDP Act 2023 · CERT-In 6-hour incident reporting · audit evidence on tap

Satellite terminals, gateways and telecommand links live inside the same zero-trust perimeter — a hacked dish must be worth nothing.

Partners, not dependencies

SPACE

IN-SPACe • NSIL / ISRO

authorisation • launch

Dhruva Space

bus • ground stations

Astrome

payload • mmWave links

Skyroot • Agnikul

future launch options

CONNECTIVITY

Jio-SES • OneWeb

leased GEO/LEO capacity

Starlink

site broadband (as available)

BSNL • BharatNet

fibre to deep districts

Skylo / NTN roaming

NB-IoT continuity

COMPUTE

Colo host (interim)

racks till our DC lives

EPC + cooling partner

DC design & build

NVIDIA OEMs

GPU supply & service

HSM / security vendors

keys, SOC tooling

RESEARCH

IIST • IITs

payload & RF talent

NAL

HAPS relay trials

C-DOT

telecom standards

Marine institutes

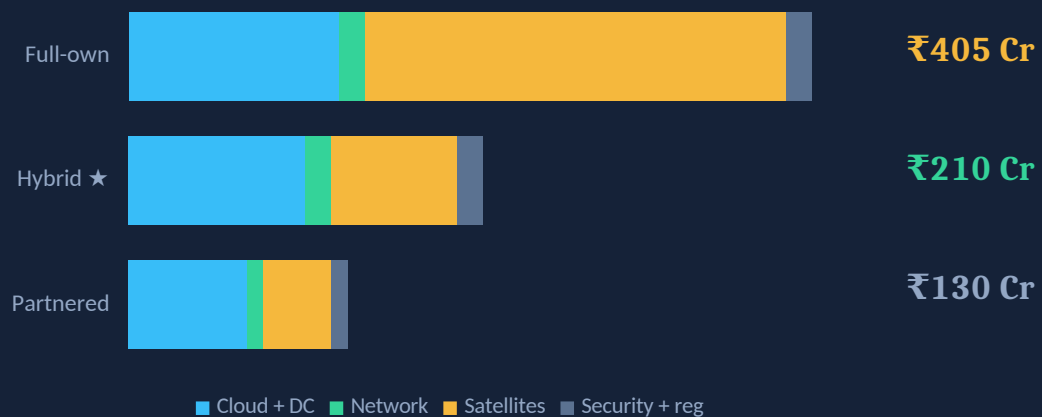
subsea DC study

The precedent that de-risks us: IN-SPACe already funded a private 12-satellite national constellation as a ₹1,150 Cr (\$135M) PPP — the collaborate path is real and priced.

Five years, three cheques, one spine

	FY27	FY28	FY29	FY30	FY31
Cloud	Private cloud, rented iron		Own GPUs in colo		Own 1 MW DC live
Network	SD-WAN + leased LEO		Captive VSAT CUG hub		200+ sites, 3-path
Satellites	File IN-SPACE + ITU		Demo build + launch		12-sat constellation

THE 5-YEAR BILL — THREE PATHS (₹ CRORE, EST.)



DECISION GATES — SPEND UNLOCKS ONLY ON PROOF

- FY28** GPU utilisation > 60% for a quarter → buy the fleet
- FY29** Demo sat healthy + licences in hand → constellation go
- FY30** Colo bill vs DC model crossover → building go

RECOMMENDED — HYBRID PATH: ₹210 Cr over 5 years